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Haoyu Zhang

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Education

M.S. **Southwest University**, Operations Research – Chongqing — Sept 2025 – present
- Top 5%(3/65): Optimization, Variational Analysis, etc.

B.S. **Donghua University**, Mathematics and Applied Mathematics – Shanghai — Sept 2021 – June 2025
- Top 14%(8/61): Mathematical Modeling, Discrete Mathematics, Data Structures and Algorithms, etc.

Knowledge & Skills

Expertise: Distributed Optimization & Primal-Dual Algorithms, Multi-agent System Modeling, Numerical Solution and Simulation of PDEs, Control System Design (PID/PI)

Languages: Mandarin (Native), English (CET-6), Japanese (JLPT N2)

Programming: Python (Pandas, PyTorch, Sklearn, OpenCV), MATLAB, Java, LaTeX/Markdown

Work Experience

Mathematical AIGC Training Intern, Appen Ltd. – Chongqing, China — Jan 2025 – Apr 2025
- Designed Chain-of-Thought (CoT) structures to improve the logical consistency and interpretability of Large Language Models (LLMs) in solving complex mathematical problems.
- Participated in establishing the company's first set of data annotation specifications for mathematical reasoning chains, achieving structured reasoning paths and unified annotation standards.

Research Experience

Fluid Dynamics and Thermal Coupling Modeling in Greenhouse Environments — Nov 2023 – Apr 2024

Group Leader
- Established 3D wind speed and temperature distribution models for glass greenhouses based on Navier-Stokes (N-S) and heat conduction equations, accounting for heat loss from plant transpiration and correcting for complex airflow in crop scenarios.
- Developed a numerical solver using finite difference and control volume methods to discretize and solve complex PDEs and visualize distribution results, providing a modeling basis for intelligent agricultural environment control.

Mathematical Modeling and Research of Mixed Human-Machine Traffic Flow — Sept 2024 – Apr 2025

Independent Researcher (Advisor: Prof. Linglong Du)
- Developed a mathematical model for mixed traffic flow of human-driven and autonomous vehicles in ring-road scenarios, incorporating the Bando-Follow the Leader (Bando-FTL) model and accounting for stochastic slowdown effects.
- Introduced a Proportional-Integral (PI) controller for Autonomous Vehicles (AVs) to optimize regulation capabilities; Matlab simulations confirmed the system effectively increased average speed and reduced variance, thereby alleviating congestion and reducing energy consumption.

Other Projects & Course Work

Climbing Posture Analysis and Coordination Optimization Based on Computer Vision, Course Project — Sept 2023 – Jan 2024
- Utilized OpenPose for precise recognition and tracking of human key points across three types of climbing scenarios.

- Designed a frame counter to calculate displacement via discrete accumulation, visualized motion curves, and computed a synchronization index through phase domain transformation to evaluate athlete coordination.

Nighttime Vehicle Speed Estimation Model Based on Optical Flow Analysis, Course Project — Mar 2024 – June 2024

- Implemented an algorithm for motion feature extraction based on Lucas-Kanade optical flow to identify vehicle contour lights.
- Analyzed the impact of optical flow thresholds on noise filtering and target recognition stability for intelligent traffic detection.

Selected Honors & Awards

First-Prize Graduate Academic Scholarship, Southwest University — 2025

Tianji Annual Scholarship, Donghua University — 2024

Honorable Mention, MCM/ICM — 2023

Third-Prize Undergraduate Academic Scholarship, Donghua University — 2022 & 2023

Third Prize (Shanghai Division), Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) — 2023

Third Prize, Asia-Pacific Mathematical Contest in Modeling (APMCM) — 2023